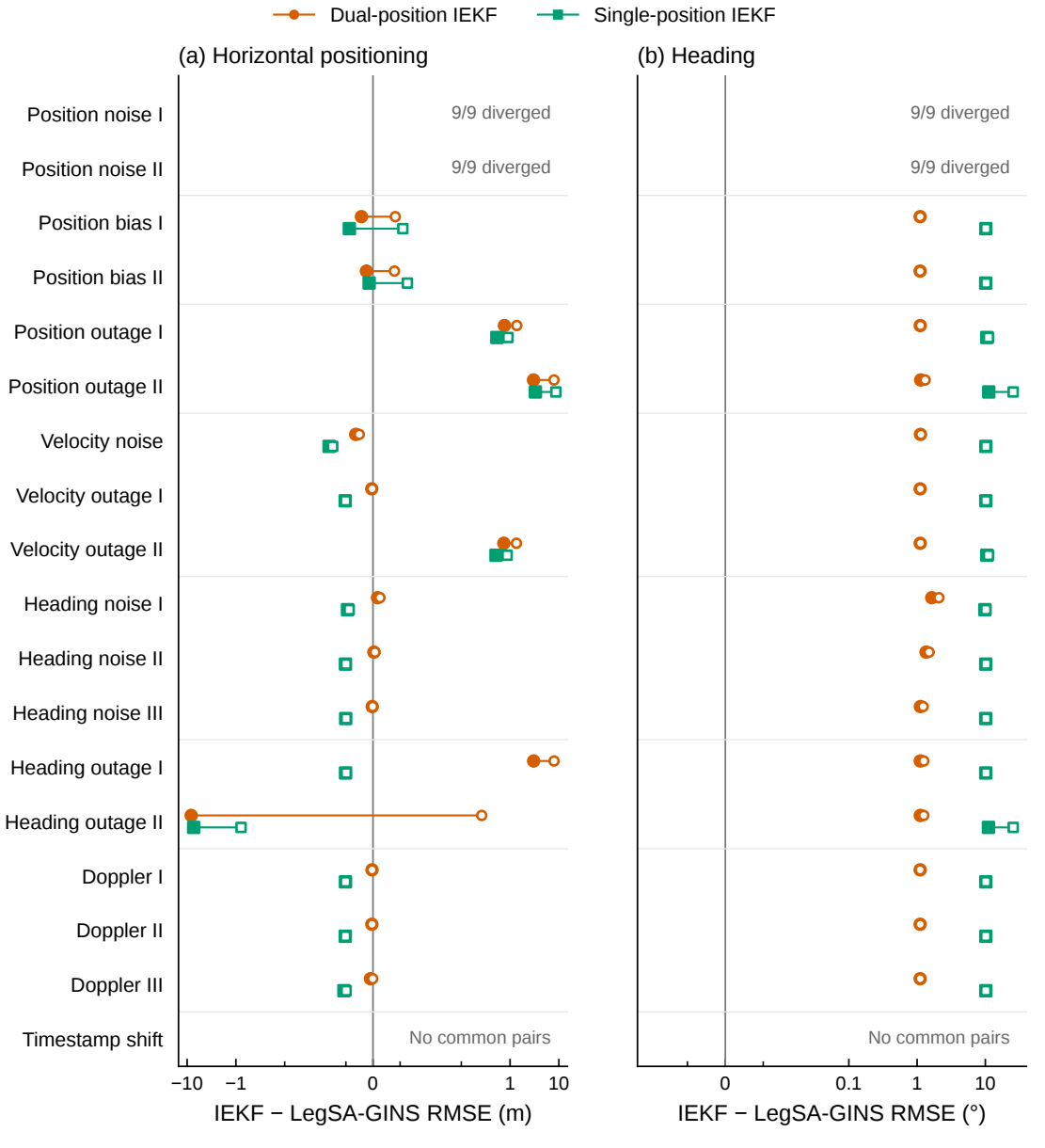
Dual-position IEKF
Single-position IEKF
(a) Horizontal positioning
(b) Heading
Position noise I
Position noise II
Position bias I
Position bias II
Position outage I
Position outage II
Velocity noise
Velocity outage I
Velocity outage II
Heading noise I
Heading noise II
Heading noise III
Heading outage I
Heading outage II
Doppler I
Doppler II
Doppler III
Timestamp shift
9/9 diverged
9/9 diverged
9/9 diverged
9/9 diverged
No common pairs
No common pairs
−10
−1
0
1
10
0
0.1
1
10
IEKF − LegSA-GINS RMSE (m)
IEKF − LegSA-GINS RMSE (°)